# Project Management Plan

**Project Title:** MataBank Banking Website
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## 1. Project Objectives

- **Automatic and Manual Wallet Setup**
    - Backend generates a unique card number, expiration date, and CVV when users create a new wallet.
    - Users can add their own existing cards manually.
- **Easy Money Management**
    - View current balance and full transaction history.
    - Deposit and withdraw funds with real-time updates.
- **Strong Security & Access Control**
    - Require users to log in before accessing wallets.
    - Enforce role-based access so only the account owner can manage each wallet.
- **User-Friendly Interface**
    - Clean, responsive design that works on desktops.
    - Immediate input checks (card number format, date, CVV) with clear error messages.
- **Reliable Performance**
    - Fast page loads (< 200 ms) and 99.9 % uptime.
    - Prevent duplicate cards or balance errors via backend checks.
- **Transaction Recording**
    - Record every deposit and withdrawal as a Transaction entry, so users, employees and admins can audit all activity.

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## 2. Project Scope

During the project development process, there will be a focus on key features and deliverables.

**In Scope:**

- Account Registration, Login, and Management
- Creation of Multiple Wallets
- Incoming Transfers Added to Wallet
- Outgoing Transfers Subtracted from Wallet
- Display Analysis of Spending Trends
- Role-Based Access Control

**Out of Scope:**

- Dedicated Matabank Mobile App (strictly a Website Platform)

## 3. Work Breakdown Structure (WBS)

| Phase | Tasks |
|---|---|
| 1. Requirements Gathering | Conduct stakeholder interviews, determine functional and non-functional requirements for a System Requirements Specification (SRS) |
| 2. Design | Begin UI/UX planning, system architecture diagrams, user flow diagrams, |
| 3. Development | Frontend and backend development, implement features like deposit, withdraw, money transfers, and security features |
| 4. Testing | Perform unit, integration, and system testing, verify requirements, confirm security measures |
| 5. Deployment | Deploy onto a secure web hosting service, and monitor launch for possible issues |
| 6. Maintenance | Track website performance, monitor for bugs, frequently update security |

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## 4. Project Schedule

| Week | Activities |
|---|---|
| 1-3 | Requirement Planning, stakeholder meetings, SRS creation |
| 4-5 | System design, wireframe models, database setup |
| 5-9 | Implementation of functional and non-functional requirements |
| 10-11 | Testing of the system and User Acceptance testing before deployment |
| 12 | Final deployment |
| 13 | Final project presentation, completed documentation |

## 5. Roles and Responsibilities

To facilitate organization, there will be key roles and responsibilities assigned to all members involved in the project development process.

| Role | Responsibilities |
|---|---|
| Project Manager | Contributes to project advancement by planning project activities, overseeing progress, and facilitating communication between all members. |
| Frontend Developer | Develops web user interfaces of the MataBank Banking Website using the applicable tools: CSS, Javascript, and HTML |
| Backend Developer | Develops logic of functions involved in money transfers, account management, and spending trend data |
| Test Engineer | Writes and executes test cases to ensure project quality while logging bugs discovered during testing. |

## 6. Risk Management Plan

| **Risk** | **Likelihood** | **Impact** | **Mitigation Strategy** |
|---|---|---|---|
| Duplicate card number collision | Medium | High | Wrap “check + create” in a database transaction with a lock, retry up to 10 times, and alert on collisions. |
| Security vulnerabilities | Low | Critical | Run automated security scans, enforce HTTPS, sanitize inputs, conduct quarterly pen tests. |
| Server downtime or slow performance | Medium | Medium | Deploy on auto-scaling servers with health checks, use caching, and maintain daily backups. |
| Requirements creep (scope changes) | Medium | Medium | Freeze scope after SRS, review all change requests at sprint boundaries and adjust schedule only by agreement. |
| Team member turnover | Medium | Medium | Cross-train team members, keep up-to-date documents, and include a 10 % time buffer for onboarding. |

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|---|---|---|---|
| Database failure or data corruption | Low | Critical | Enable automated daily backups, use a replicated database setup, and test restores regularly. |
| Browser compatibility issues | Low | Low | Test UI on major browsers and devices, use a CSS framework with builtin cross-browser support, and fix issues early. |
| Overlapping transaction errors | Medium | High | Use database transactions and row-level locks for deposit/withdraw operations, and add retry logic on conflicts. |

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## 7. Communication Plan

| Activity | Audience | Frequency | Format |
|---|---|---|---|
| Daily Stand-up | Dev Team | Every Week | 15 min Zoom or Call |
| Weekly Status Update | PM & Stakeholders | Every Week | Email summary + project dashboard link |
| Monthly Progress Report | PM & Stakeholders | Every month | PDF report + dashboard link |
| Show New Features | Dev Team, PM, Stakeholders | Every 2 weeks | Live demo + quick feedback session |
| Code Review | Dev Team | Every Week | GitHub pull-request reviews |
| Track & Solve Issues | Dev & QA Team | Ongoing | Slack channel + Jira tickets |
| Reflect & Improve | Dev Team & QA | End of each 2-week cycle | 30-min meeting to discuss lessons learned |
| Design Review | Dev Team, Designer, PM | Every month | Video call with screen share |
| Security Check-in | Security Lead & PM | Every month | 30-min call + short security report |

| | | | |
|---|---|---|---|
| Release Announcement | All Users & Stakeholders | With each release | Email blast + in-app notification |
| Urgent Alerts | Dev, QA, PM | If needed | Slack ping + short Zoom call |
| Final Presentation | All Stakeholders | End of project | Live demo + overall summary report |

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## 8. Tools and Technologies

**Development**:

- **VSCode**: code editor IDE
- **Web Forums**: for specific problem solving
- **MySQL**: Database
- **Javascript**: Front and Backends
- **HTML**: Front end
- **CSS**: Front End
- **Node.Js**: Backend

**Communication**:

- **Discord**: main source of communication
- **Email**: extra communication (sharing docs)
- **Github**: sharing code between each other
- **Flash drive**: sharing code between each other

**Testing**:

- **Jest or Mocha**: used for unit testing JavaScript functions and modules
- **Supertest**: for testing API endpoints during integration testing
- **Cypress**: to automate full workflows like login, fund transfer, and viewing transaction history
- **OWASP ZAP (light mode)**: to perform lightweight scans for common web vulnerabilities
- **Lighthouse (Chrome DevTools)**: to evaluate site speed, accessibility, and performance
- **Uptime Robot:** to evaluate uptime.

## 9. Success Criteria

### Unit and Integration Testing

- **Criterion**: At least 95% of unit and integration test cases pass.
- **Measurement**: Automated testing reports from Jest and SuperTest
- **Goal**: Ensure each component and its integrations behave as expected in isolation and in combination.

### System and End-to-End Testing

- **Criterion**: At least 90% of system and E2E test cases pass, with 0 critical or high-priority defects open.
- **Measurement**: Test case execution results from Cypress.
- **Goal**: Validate the full functionality of the banking workflow (e.g., login, account management, transactions).

### Security Testing

- **Criterion**: No critical or high-severity vulnerabilities detected in the final security audit.
- **Measurement**: Reports from penetration testing, using OWASP ZAP.
- **Goal**: Ensure the system is resistant to common threats like XSS, SQL injection, and CSRF.

### Performance Testing

- **Criterion**: System handles at least 1000 concurrent users with response time < 2 seconds for 95% of requests
- **Measurement**: Load test reports using Lighthouse.
- **Goal**: Ensure site responsiveness and scalability under load.

### Uptime and Availability

- **Criterion**: Website maintains 99.9% uptime during the first 30 days post-launch.
- **Measurement**: Monitoring logs from tools like UptimeRobot
- **Goal**: Validate operational stability of the deployed system.